THE NOVA RESEARCH CURRICULUM

Biology

Grade 7 · Term 2

Student edition

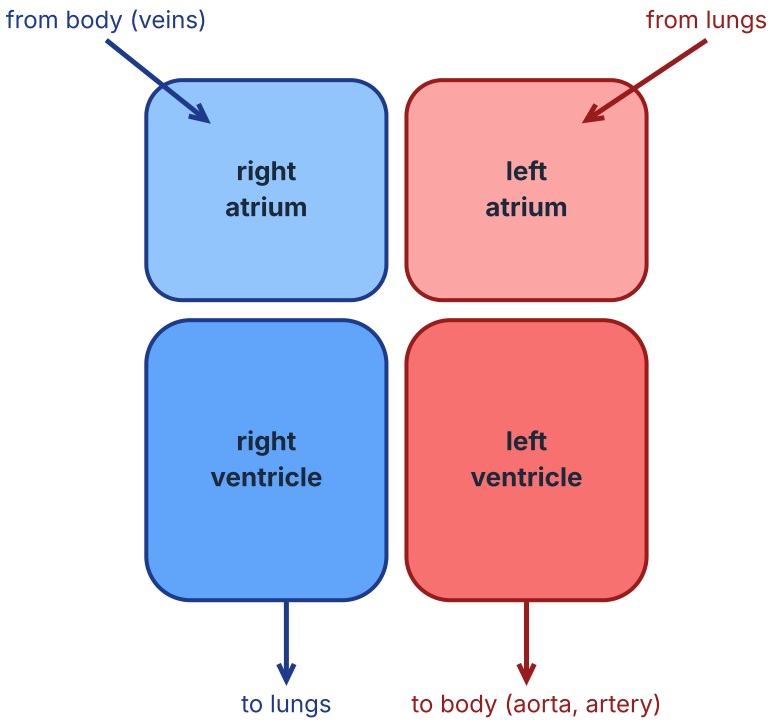
Outcome: BI-7.2 respiratory and circulatory systems; aerobic and anaerobic respiration.

Lesson notes

- **Respiratory system (Weeks 1–4):** nose, trachea, bronchi, lungs, alveoli. Gas exchange in the alveoli: oxygen into the blood, carbon dioxide out. Breathing: diaphragm and rib muscles. Smoking damages the alveoli. Home model with a bottle and balloons.

  ▶ **Try it on NovaSchool:** Breathing model: lungs and diaphragm — open **Simulations** on your dashboard.

- **Circulatory system (Weeks 5–8):** the heart has four chambers and pumps blood in a double circulation (to the lungs and to the body). Arteries carry blood away from the heart; veins carry it back; capillaries link them. Blood: red cells (carry oxygen), white cells (fight germs), platelets (clotting), plasma. Measure pulse before and after exercise.



The view is drawn as if you face the person, so their left side is on your right.

The four chambers of the heart and the double circulation (simplified).

- **Respiration (Weeks 9–12):** aerobic: glucose + oxygen → carbon dioxide + water + energy. Anaerobic in muscles during hard exercise: glucose → lactic acid + a little energy (cramps). In yeast: glucose → alcohol + carbon dioxide + energy (bread rising). Respiration is not the same as breathing.

Workbook pages

Page	Task
1	Label the respiratory system.
2	Label the heart; trace the path of blood.
3	Pulse practical: table and bar chart.
4	Compare aerobic and anaerobic respiration.

Question bank

#	Outcome	Question	Your answer
1	BI-7.2	Gas exchange happens in the ...	trachea / alveoli / heart / stomach
2	BI-7.2	How many chambers does the human heart have?	2 / 3 / 4 / 6
3	BI-7.2	Arteries carry blood ... the heart.	to / away from / around / inside
4	BI-7.2	Which blood cells carry oxygen?	white cells / platelets / red cells / plasma
5	BI-7.2	Anaerobic respiration in muscles makes ...	alcohol / lactic acid / oxygen / starch
6	BI-7.2	Breathing and respiration are the same thing. True or false.	True / False
7	BI-7.2	Why does your pulse rise during exercise?	_____
8	BI-7.2	Yeast makes bread rise by releasing carbon dioxide. True or false.	True / False